

WHEN DOES FEATURE GEOMETRY HELP DATA-POOL SCREENING?

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ABSTRACT

Can unlabeled feature geometry reduce the number of data-pool combinations requiring supervised evaluation? We study screening with frozen representations, separating spectral information, conditional risk models, and empirical utility. A constructive example shows that marginal singular spectra and principal angles do not determine merged effective rank. A standard Bayesian linear model motivates target-weighted A-optimal screening, but its risk identity does not extend to arbitrary conditional shift or softmax classification. We compare it with target-blind diversity and same-information second-moment matching. On PACS and Office-Home, A-optimal screening fails a prespecified 2% relative Brier improvement threshold; exhaustive audits reveal only 0.781% mean oracle headroom over the strongest target-blind baseline. In a subsequent equal-cost DomainNet study, 455 combinations per target each contain 1,536 source samples. Shortlisting 227 combinations retains 98.333% of the independent-test top ten on average across six target domains and two encoders. However, second-moment MMD attains the same primary recall and all twelve validation-selected combinations. Moreover, the entire within-task relative Brier span is below 0.425%, making a 5% regret threshold uninformative. These results support a bounded role for target-conditioned second-order geometry in pre-screening, not a superior A-optimal selector or a universal label-free utility law. We provide auditable manifests, rankings, and exhaustive outcome tables.

1 INTRODUCTION

Selecting source data with frozen encoders is an appealing alternative to training every candidate model. A collection of feature matrices already exposes sample counts, singular spectra, and cross-pool geometry without requiring a supervised fit. The unresolved question is what these quantities justify: predicting the spectrum of a merged representation, predicting a downstream risk, or merely preserving useful candidates for later evaluation. These are different objectives, and success at one need not imply success at another.

We study *data-pool screening*: rank a finite collection of source-pool combinations using unlabeled source features and, when permitted, an unlabeled target sample; retain a shortlist; then use supervised validation to choose its final member. This setting is deliberately narrower than fully label-free data selection. Labels are hidden from the screening rule, not removed from the subsequent training and validation procedure. Our aim is to determine when second-order geometry is informative enough to reduce this evaluation search space.

The distinction matters mathematically. Full feature Gram operators determine a merged singular spectrum, whereas marginal spectra and principal angles can omit how singular mass aligns with shared directions. Even exact merged geometry does not identify a supervised target task. A shared Bayesian linear model supplies a conditional bridge to target-weighted A-optimal design, but does not establish a general connection to classification utility. We use this bridge to motivate an empirical comparator, not to relabel a standard optimal-design criterion as a new theorem or a universally reliable selector.

The empirical picture is similarly conditional. On PACS and Office-Home, the target-weighted criterion has a small Brier advantage but fails the prespecified practical-effect threshold. An exhaustive, explicitly post-hoc audit finds little average room for any selector to improve over a strong

target-blind comparator. A subsequent DomainNet experiment fixes the number of source samples, broadens the candidate construction, and separates validation from official test data. Here, screening half of the combinations retains almost all of the test top ten. Yet a simpler second-moment matching baseline obtains the same primary result, and the absolute Brier differences are small enough to make a generous regret threshold vacuous.

Our contribution is an analysis of these distinctions and their consequences. First, we give a constructive spectral insufficiency example and state the assumptions needed for a risk interpretation. Second, we evaluate target-conditioned and target-blind scores under an access-controlled, equal-cost shortlist protocol with exhaustive outcome audits. Third, we expose three diagnostics that qualify apparent success: oracle headroom, metric resolution, and the difference between score certificates and useful decisions. The evidence supports a bounded screening role for target-conditioned geometry, not a new state-of-the-art selection method.

2 PROBLEM AND RELATED WORK

Let $H_i \in \mathbb{R}^{n_i \times d}$ be the features of source pool i , produced by a fixed encoder and a declared, label-independent preprocessing map. Write $G_i = H_i^\top H_i$. A feasible combination S belongs to a finite family $\mathcal{C}$; $G_S = \sum_{i \in S} G_i$ and $n_S = \sum_{i \in S} n_i$. An unlabeled target-selection sample estimates the uncentered second moment $C_T = \mathbb{E}_T[hh^\top]$. A screen produces $\mathcal{Q}_s \subset \mathcal{C}$ with $|\mathcal{Q}_s| = s$. A separate target-validation set chooses $\hat{S} \in \mathcal{Q}_s$ after source-supervised training. Target-test data are reserved for auditing this frozen choice and the shortlist.

We distinguish two access levels. At U0, a rule sees source features and target-selection features but neither source labels nor target validation/test data. At U2, it may fit on source labels and select using target-validation labels. A target-blind rule uses less information than U0. Comparing against both types of rules separates the value of target information from the value of the particular functional applied to it. Label use in defining the benchmark taxonomy is disclosed separately from selector access.

Spectral diagnostics. Effective-rank criteria such as RankMe assess learned representations without downstream labels (Garrido et al., 2023). This motivates a diversity comparator, but selecting among encoders is not the same problem as selecting combinations of pools with one frozen encoder. In our setting, even a perfect merged-rank predictor need not rank supervised risks correctly. We therefore keep spectral fidelity and downstream usefulness as separate endpoints.

Optimal design and moment matching. Trace-of-covariance objectives are standard in A-optimal design and active linear regression (Fontaine et al., 2021). Target weighting changes which prediction directions matter; it does not make the underlying posterior covariance identity novel. Second-moment matching is a direct instance of empirical MMD with a homogeneous degree-two polynomial kernel (Gretton et al., 2012). Moment alignment also appears in multi-source domain adaptation (Peng et al., 2019). Our comparison freezes representations and ranks pool combinations; it neither retrains a domain-invariant representation nor claims a new MMD construction.

Diversity and cost. Determinantal objectives favor diverse subsets (Kulesza & Taskar, 2012). We implement a declared subspace-kernel comparator, not every possible DPP or coresets algorithm. Compute-constrained selection must account for both selection and subsequent training (Yin & Rush, 2025). Accordingly, a reduction in evaluated combinations is reported as such, rather than converted into an unmeasured end-to-end speedup. The contribution is the matched audit and its boundaries, not a claim that classical alternatives are exhausted.

3 WHAT GEOMETRY CAN AND CANNOT PREDICT

3.1 EXACT COMPOSITION REQUIRES DIRECTIONAL INFORMATION

For vertically stacked features with fixed nonnegative source weights w_i ,

$$H_S = \begin{bmatrix} \sqrt{w_{i_1}} H_{i_1} \\ \vdots \\ \sqrt{w_{i_k}} H_{i_k} \end{bmatrix}, \quad H_S^\top H_S = \sum_{i \in S} w_i G_i. \quad (1)$$

Thus the full Gram sum exactly determines the merged singular values. This identity presumes the same feature coordinates and fixed preprocessing. Re-centering each proposed merge or re-fitting a projection for each combination defines a different problem. Our reported effective rank uses singular values, not squared singular values:

$$r_{\text{eff}}(H) = \exp \left(- \sum_j p_j \log p_j \right), \quad p_j = \frac{\sigma_j(H)}{\sum_k \sigma_k(H)}. \quad (2)$$

For a nonzero matrix, zero terms use $0 \log 0 = 0$. Numerical implementations apply a declared singular-value rank tolerance (Appendix B); it is not part of the exact identity.

Proposition 1 (Marginal spectra and principal angles are insufficient). *There exist row-normalized pairs (A, B_1) and (A, B_2) with identical marginal singular spectra and identical principal angles between their row spaces, but different effective ranks after vertical concatenation.*

Proof. Take A with four rows $e_1^\top$ and one row $e_2^\top$. Let B_1 have four rows $e_1^\top$ and one $e_3^\top$, and let B_2 have one row $e_1^\top$ and four $e_3^\top$. All three nonzero singular spectra are $(2, 1)$; both row-space pairs have principal angles $(0, \pi/2)$. The merged Gram spectra are respectively $(8, 1, 1)$ and $(5, 4, 1)$. Equation (2) yields approximately 2.6260 and 2.8495. $\square$

The missing information is the assignment of singular mass to the shared versus private directions. Under the additional assumption of mutually orthogonal, paired two-dimensional invariant blocks, a block with singular weights a, b and overlap c has eigenvalues

$$\lambda_{\pm} = \frac{a^2 + b^2 \pm \sqrt{(a^2 - b^2)^2 + 4a^2b^2c^2}}{2}. \quad (3)$$

This is an exact local calculation, not a general reduction to four scalar summaries. Appendix A derives it and explains its scope.

3.2 A RISK INTERPRETATION NEEDS A SHARED CONDITIONAL MODEL

Full source and target feature geometry still cannot identify arbitrary target utility. For example, let every feature equal 1, and let two source pools have constant labels $+1$ and -1 . Two target worlds can have identical features and opposite constant labels. Source-trained constant predictors exchange their target-risk ordering between these worlds. Any feature-only screen sees the same input in both. Additional assumptions or labeled evidence are therefore necessary for a universally correct preference.

A useful conditional model is scalar Bayesian linear regression. Assume $\theta \sim N(0, P_0)$, $P_0 \succ 0$, and $y = h^\top \theta + \epsilon$ with independent $\epsilon \sim N(0, \sigma^2)$, $\sigma^2 > 0$. The same θ and noise model govern all sources and the target. Conditional on the source design, the posterior covariance and posterior-mean prediction risk are

$$P_S = (P_0^{-1} + \sigma^{-2} G_S)^{-1}, \quad R_T(S) = \sigma^2 + \text{tr}(C_T P_S). \quad (4)$$

The expectation integrates posterior uncertainty and a new target observation. This is a standard conjugate calculation, recalled in Appendix A; it is not a distribution-free risk bound. It motivates ranking combinations by $f_A(S) = \text{tr}(\hat{C}_T P_S)$ with fixed hyperparameters. Adding a PSD source Gram cannot increase this score, but we make no general diminishing-returns or greedy global-approximation claim.

The same-information alternative is

$$f_M(S) = \left\| G_S/n_S - \hat{C}_T \right\|_F^2. \quad (5)$$

This is squared distance between empirical second-moment embeddings for $k(h, h') = (h^\top h')^2$, including diagonal terms. It is uncentered moment matching, not the centered covariance loss sometimes used in adaptation. Both scores see the same unlabeled data. Their different operations on that information must earn any relative advantage empirically.

The real-data readout is not the generative model in Equation (4). We fit one-hot ridge regression and apply softmax before computing multiclass Brier loss. Conditional shift, model misspecification, and this nonlinear transformation all break an automatic identification of f_A with the evaluated loss. Consequently, agreement in the synthetic shared-model setting tests the conditional motivation, whereas success on classification is an empirical observation.

3.3 SCREENING AND VALIDATION HAVE DIFFERENT ERROR SOURCES

Let $\ell(S)$ be a target risk for fixed trained predictors, and let $v(S)$ be their validation losses. Suppose $|v(S) - \ell(S)| \leq \varepsilon$ for every $S \in \mathcal{Q}_s$. For $\hat{S} \in \arg \min_{S \in \mathcal{Q}_s} v(S)$,

$$\ell(\hat{S}) - \min_{S \in \mathcal{C}} \ell(S) \leq \underbrace{\min_{S \in \mathcal{Q}_s} \ell(S) - \min_{S \in \mathcal{C}} \ell(S)}_{\text{shortlist omission}} + 2\varepsilon. \quad (6)$$

This elementary deterministic inequality separates missing a good candidate from selecting badly within the shortlist. A proof is included in Appendix A. The experiments audit the finite-test counterparts of these quantities; they do not establish a population uniform-deviation value ε . Recall of the test top ten further assesses whether a screen preserves multiple good candidates, rather than merely one observed oracle.

4 EVALUATION PROTOCOL

Study sequence. We first test the risk motivation with controlled synthetic tasks, then evaluate one-shot selection on PACS and Office-Home (Li et al., 2017; Venkateswara et al., 2017). Their failed practical-effect tests are retained. An exhaustive oracle-headroom analysis is performed after opening those outcomes and is labeled post hoc. DomainNet is a subsequent exploratory hypothesis on a different dataset, with its own configuration frozen before feature extraction and outcome evaluation. It is not a retroactive confirmation of the earlier hypothesis. Appendix B maps these studies to immutable result directories.

Fixed-cost DomainNet candidates. We use the official cleaned DomainNet data (Peng et al., 2019), selecting 100 class names by a fixed hash rule. In each of six domains, disjoint roles contain 1,536 target-selection images, 1,024 target-validation images, and 3,072 anchor-pool images from official training data, plus 2,048 images from official test data. The manifest therefore contains 46,080 samples. Benchmark construction uses class metadata; screening does not use their labels.

CLIP ViT-B/32 (Radford et al., 2021) is used only as a frozen construction anchor. Within each domain, its first principal-component scores define low, middle, and high strata; each candidate block contains 512 images. For any target domain, we exclude its three blocks and enumerate $\binom{15}{3} = 455$ combinations. Every combination contains exactly 1,536 source samples. ResNet-50 and DINOv2-B/14 provide the evaluation features (He et al., 2016; Oquab et al., 2024). Row normalization, a fixed Rademacher projection to 128 dimensions, and post-projection row normalization are shared by all rules. The construction encoder is not an evaluation encoder, but this alone does not establish independence from pretrained model corpora.

Scores and baselines. Target A-opt uses $P_0^{-1} = I$ and $\sigma^2 = 1$. We compare second-moment MMD, target energy $\text{tr}(\hat{C}_T G_S/n_S)$, Bayesian D-opt, merged effective rank, subspace DPP, domain balance, and 20 seeded random rankings. All rules score the same 455 combinations; there is no unequal search budget in this experiment. Domain balance additionally uses source-domain identifiers,

not target outcome labels. Appendix B specifies the DPP kernel and tie rules. Classical coverage and moment baselines are controls with explicit implementations, not straw-man labels for unspecified algorithms.

Screen, validate, freeze, audit. The screen emits immutable rankings without reading target-validation or target-test features or labels. It retains nested shortlists of sizes 10, 25, 50, 100, 227, with 227 fixed as primary. The U2 stage independently evaluates each method’s 227 combinations, fits ridge with regularization 1 and no intercept, and chooses minimum validation Brier with lexicographic tie-breaking. Smaller shortlist outcomes reuse only that method’s nested evaluations. No cross-method result cache is used. The test stage verifies upstream hashes before exhaustively evaluating all 455 combinations. This expensive oracle audit is a research cost, not an action available to deployment.

For test Brier $b(S)$ and its exhaustive minimizer S^* , we report

$$\text{Recall}_{10}(\mathcal{Q}_s) = \frac{|\mathcal{Q}_s \cap \text{Top}_{10}(b)|}{10}, \quad \text{Regret}(\hat{S}) = \frac{b(\hat{S}) - b(S^*)}{|b(S^*)|}. \quad (7)$$

The denominator is the oracle loss, *not* the best-to-worst span. The frozen primary gate requires at least 50% removal, mean recall at least 90%, and mean normalized regret at most 5%. Brier is primary; accuracy, squared loss, NLL, macro-F1, and calibration error are supporting outcomes. They are not interchangeable with the primary metric.

Units and uncertainty. For real data we first average encoders within each dataset–target-domain unit; random repetitions are averaged within task. PACS/Office-Home have eight such units and DomainNet six. Bootstrap intervals resample those units. Neither 455 overlapping combinations nor two encoders create hundreds of independent tasks. Even target-domain units share source pools, so these intervals are descriptive stability summaries under that resampling scheme, not evidence of broad independent-dataset generalization.

5 EMPIRICAL FINDINGS

5.1 THE RISK MOTIVATION WORKS UNDER ITS ASSUMPTIONS, NOT UNDER ARBITRARY SHIFT

The corrected synthetic sweep uses 20 independent generation seeds and 4,320 configurations, varying feature dimension, candidate count, block budget, target sample size, and target rank. Within each setting, target-sample generation is controlled across comparisons. Under the shared conditional model, target A-opt reduces modeled risk by 4.23% relative to the strongest target-blind effective-rank comparator (95% interval [3.86, 4.62]%) and by 14.85% relative to second-moment MMD ([13.99, 15.70]%). These intervals bootstrap 216 design settings after averaging their 20 seeds; they describe that fixed design grid.

These are conditional synthetic gains. In a separate source-conditional-shift sweep, its mean normalized prediction regret increases from 0.211 at shift 0 to 2.197 and 3.143 at shifts 0.5 and 1. The shift-free value differs from the posterior-risk experiment because this sweep evaluates a different, realized prediction endpoint. The deterioration is consistent with a violated shared-mechanism assumption; it is not a refutation of the correctly conditioned covariance identity. Nor do numerous parameter-grid rows constitute numerous independently sampled real domains.

5.2 SMALL REAL-DATA GAINS COINCIDE WITH LIMITED ORACLE HEADROOM

On PACS and Office-Home at block budget $K = 3$, target A-opt has mean Brier 0.826769, versus 0.829162 for DPP and 0.828719 for MMD. The mean of within-unit relative improvements is 0.351% over DPP and 0.273% over MMD. Against DPP the paired Brier difference interval is [−0.007449, 0.000731]; against MMD it is [−0.003392, −0.000712]. The latter is directionally consistent but still far below the frozen 2% practical-effect requirement. Both primary gates fail. These comparisons hold the number of blocks fixed, not the number of source samples.

Table 1: DomainNet, $s = 227$. Means average two encoders within each of six target domains; random repetitions are not independent units. Regret is a percentage of oracle Brier, not of the candidate span. All methods pass the loose regret sub-gate; only A-opt and MMD pass the recall sub-gate.

Screen	Recall (%)	Regret (%)	Spearman
Target A-opt	98.333	0.001311	0.7443
Second-moment MMD	98.333	0.001311	0.6902
Bayesian D-opt	82.500	0.002658	0.4996
Merged effective rank	81.667	0.002909	0.4951
Random (20 repeats)	48.958	0.003899	0.0016
Target energy	46.667	0.009347	-0.0399
DPP subspace	27.500	0.009067	-0.0633
Domain balance	25.000	0.012083	0.0681

The subsequent exhaustive audit evaluates all 220 $K = 3$ combinations per task. Relative to DPP, the oracle offers only 0.781% mean headroom, with descriptive 95% interval $[0.162, 1.617]\%$. Only PACS photo exceeds 2% at the target-unit level (Figure 2, left). The oracle’s mean remaining gain over the post-hoc envelope of the tested parent methods is 0.087%. This envelope uses test outcomes and is not a deployable selector. Low headroom helps explain why large average gains are unavailable in this candidate family; it does not turn the failed original tests into successes.

5.3 EQUAL-COST SCREENING PRESERVES GOOD CANDIDATES, WITH MMD PARITY

Table 1 gives the DomainNet primary result. Retaining 227 of 455 combinations removes 50.110% and gives target A-opt 98.333% mean test top-ten recall. Its descriptive recall interval is $[96.667, 100]\%$ and normalized selected regret is 0.001311%, with interval $[0.000228, 0.003031]\%$. Every primary shortlist contains a test Brier oracle, so the observed omission term is zero across all twelve encoder–target tasks. The small remaining regret comes from the validation-selected choice within the shortlist.

Second-moment MMD obtains exactly the same primary recall and selected regret. Moreover, validation selects the same final combination for both methods in all twelve tasks. Their mean primary-shortlist Jaccard similarity is 0.8307 in a post-hoc comparison. A-opt has higher full-ranking Spearman correlation with test Brier (0.7443 versus 0.6902), but MMD has higher mean top-ten recall at all four smaller shortlist sizes (Figure 1). Better global correlation therefore does not imply a better operating point for screening.

This is evidence that target-conditioned second-order information is useful for this shortlist protocol. It is not evidence that inverse-covariance weighting is necessary: direct moment matching is competitive under identical feature access. It also does not show that added candidate heterogeneity caused the improvement relative to PACS/Office-Home. Dataset, projection dimension, sample budget, and candidate construction changed together; no isolated heterogeneity intervention was performed.

5.4 LOW REGRET AND VALID INTERVALS CAN BE UNINFORMATIVE

In a post-hoc diagnostic, the full relative Brier span $[\max_S b(S) - \min_S b(S)] / \min_S b(S)$ ranges from 0.0280% to 0.4242% across the twelve DomainNet tasks. Thus *every* possible candidate satisfies the 5% regret threshold, regardless of screening. Passing that sub-gate supplies no discrimination here. The useful primary evidence is preservation of high-ranked candidates at a fixed evaluation count, with the important caveat that exact top-ten membership can itself be sensitive to small finite-test differences.

Accuracy varies across combinations and its oracle does not always coincide with the Brier oracle. This cautions against describing low Brier regret as general classification-performance preservation. Because the readout applies fixed-temperature softmax to one-hot ridge scores, calibration and score

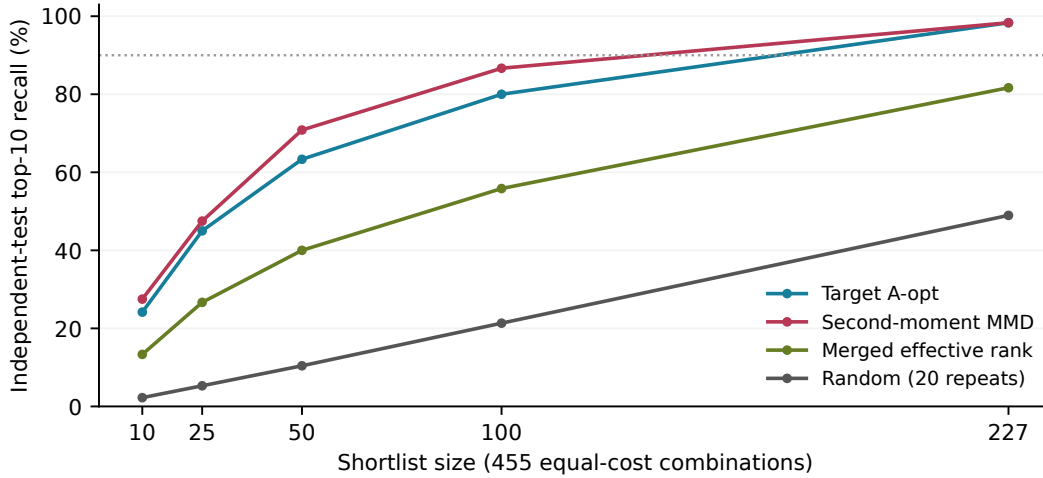


Figure 1: DomainNet shortlist-size curves from the frozen outcome tables. The dotted line is the prespecified 90% recall threshold. MMD has higher mean recall at smaller shortlists and ties A-opt at the primary size. Curves are descriptive means, not tests of pairwise significance.

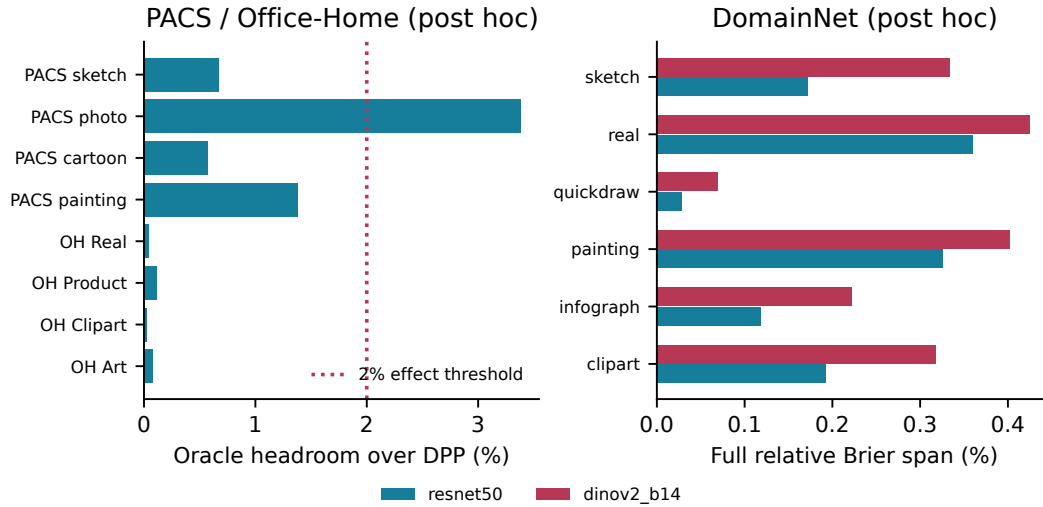


Figure 2: Post-hoc diagnostics, without changing frozen gates. Left: relative oracle improvement over DPP for eight PACS/Office-Home target units, averaging encoders first. Right: within-task best-to-worst relative Brier span on DomainNet; all bars are below 0.425%, far below the 5% regret gate. These quantities answer different questions and are not directly comparable effect sizes.

scale are plausible contributors to small Brier differences, but the current experiment does not establish that mechanism. Repeated independent tests or per-example paired resampling would be needed to assess top-ten stability; neither is claimed here.

Compression certificates show a related distinction. In the earlier four-encoder spectral rank sweep, all 2,880 selected-prefix intervals cover their exact scores, yet no greedy step is certified among 7,200 steps at sketch ranks at most 50. A separate target-risk sketch study over 20 seeds and 1,440 configurations also fails its byte-budget gate: among full-grid strategies using at most one quarter of packed full-Gram bytes on average, the best selected-set agreement with Full-Gram greedy is 15.42%, with a 3.91% pairwise certificate rate. Coverage is a correctness check, not sufficient evidence of decision utility or practical communication savings (Appendix C).

6 DISCUSSION AND CONCLUSION

The evidence separates three claims. Gram additivity and compatible local spectral calculations are exact algebraic statements. Target-weighted A-optimal risk is justified by a shared Bayesian linear model. Useful shortlist screening in real classification remains an empirical property of a candidate family, readout, target sample, and operating point. None of these implications can be freely reversed or extended.

For the audited equal-cost DomainNet construction, second-order target geometry preserves substantially more test top-ten candidates than the tested target-blind diversity rules at half the evaluation count. The simplest successful same-information rule, second-moment MMD, is equally important to this conclusion as A-opt. PACS/Office-Home failures, limited oracle headroom, loose regret gates, and weak compression decisions prevent a broader utility or superiority claim.

The main limitations are one new equal-cost dataset, six correlated target units, one frozen projection seed, one linear readout, and no independent demonstration of end-to-end time savings. The next evidentiary steps are an independent fixed-cost dataset, stability of the chosen metric and top-ten set, matched total-cost measurements, and a controlled test of when inverse-covariance weighting adds value over moment matching. Until those tests are completed, feature geometry is best understood as a conditional pre-screening signal that requires subsequent labeled validation and explicit failure diagnostics.

AI USE STATEMENT

Generative AI tools assisted conceptual reformulation, mathematical claim and proof drafting, experiment design, software implementation, interpretation of results, literature search, manuscript writing, figure scripting, and Chinese translation. Numerical findings in this draft are taken from recorded experiment artifacts rather than generated observations. AI assistance is not independent validation of those artifacts or of the mathematical claims. Final verification and responsibility for a submitted version remain with the human authors.

REPRODUCIBILITY STATEMENT

Proof details appear in Appendix A; preprocessing, access rules, candidate construction, metrics, and result paths appear in Section 4 and Appendix B. The code repository is <https://anonymous.4open.science/r/collapsemodel-1DB6>. Outcome tables and manifests support aggregate reproduction; large feature caches are identified by hashes rather than duplicated in Git. Exact hardware-independent bitwise feature reproduction is not claimed.

ETHICS STATEMENT

The study uses existing benchmark images and pretrained encoders, with no new participant recruitment. Public availability does not remove image licensing, privacy, or representational-bias concerns. We do not claim that a diverse feature spectrum or a high shortlist recall makes a selected pool socially representative, fair, or private. The candidate taxonomy, pretraining overlap uncertainty, and restriction to visual classification limit downstream deployment claims. Dataset and model access remain subject to their original terms.

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A PROOFS AND SCOPE OF THE MATHEMATICAL CLAIMS

Local two-direction calculation. Let u, v be unit directions with $u^\top v = c \in [0, 1]$. In the orthonormal basis u, z , where $v = cu + \sqrt{1 - c^2}z$, the operator $a^2uu^\top + b^2vv^\top$ is

$$\begin{pmatrix} a^2 + b^2c^2 & b^2c\sqrt{1 - c^2} \\ b^2c\sqrt{1 - c^2} & b^2(1 - c^2) \end{pmatrix}. \quad (8)$$

Its trace is $a^2 + b^2$ and determinant $a^2b^2(1 - c^2)$, yielding Equation (3). At $c = 1$, interpret the second direction as a zero branch. A collection of these eigenvalue pairs gives a full spectrum only when the associated blocks are mutually orthogonal and invariant for both source Grams; unpaired directions contribute their own eigenvalues. Principal vectors alone need not be singular vectors. The row-normalized counterexample in the main text already violates the inference that marginal spectra and principal-angle lists suffice, without any complication from different sample counts.

Posterior risk identity. For fixed H_S and Gaussian likelihood, completing the square in the prior times likelihood gives posterior precision $P_0^{-1} + \sigma^{-2}H_S^\top H_S$ and mean $\mu_S = P_S\sigma^{-2}H_S^\top y_S$. For an independent target observation $y = h^\top \theta + \epsilon$, conditional on the training data and h ,

$$\mathbb{E}[(y - h^\top \mu_S)^2 \mid H_S, y_S, h] = \sigma^2 + h^\top P_S h. \quad (9)$$

Integrating h gives Equation (4). We assume the target covariate distribution does not reveal additional information about θ beyond this model. If a source has a different regression function, or labels are multiclass with softmax readout, this derivation no longer identifies the evaluated loss. The proof does not bound this misspecification error. Replacing C_T by a finite-sample estimate adds another discrepancy not removed by the identity.

Monotonicity and a deterministic sketch interval. If $G \succeq \tilde{G} \succeq 0$, inverse order implies $(P_0^{-1} + \sigma^{-2}G)^{-1} \preceq (P_0^{-1} + \sigma^{-2}\tilde{G})^{-1}$. Taking a trace against a PSD target moment preserves this order. More generally, suppose the sum of omitted sketch tails satisfies $0 \preceq G - \tilde{G} \preceq \delta I$. Define $B = P_0^{-1} + \sigma^{-2}\tilde{G}$. Then

$$\text{tr}(\hat{C}_T(B + \sigma^{-2}\delta I)^{-1}) \leq f_A(G) \leq \text{tr}(\hat{C}_T B^{-1}). \quad (10)$$

One admissible δ is the sum of valid spectral-norm bounds on each PSD tail. This elementary interval illustrates how a valid score bound can be too broad to determine an ordering. Certifying one candidate requires its upper bound to be below every competing lower bound. It certifies only that comparison of the stated proxy, not a global optimum, label privacy, or true classification risk. The archived sketch sweep uses its documented implementation and byte accounting; Equation (10) is not a claim that all possible tighter bounds were evaluated.

Validation decomposition. Let S_Q minimize ℓ over $\mathcal{Q}_s$. Uniform deviation on that shortlist gives

$$\ell(\hat{S}) \leq v(\hat{S}) + \varepsilon \leq v(S_Q) + \varepsilon \leq \ell(S_Q) + 2\varepsilon. \quad (11)$$

Subtracting $\min_{S \in \mathcal{C}} \ell(S)$ proves Equation (6). No probability is asserted by this algebra. For population generalization, one must additionally justify the deviation event using the actual dependence and selection scheme. Exhaustive finite-test scores alone do not supply that justification.

Polynomial-kernel MMD. The feature map $\phi(h) = hh^\top$, equipped with Frobenius inner product, satisfies $\langle \phi(h), \phi(h') \rangle_F = (h^\top h')^2$. The squared distance between source and target empirical feature means is therefore Equation (5). This is the biased empirical squared MMD (a V-statistic), not the diagonal-excluded unbiased estimator. It depends on uncentered second moments and cannot distinguish all distributions sharing those moments. We use it as a computationally simple same-information comparator, not a complete discrepancy measure.

B IMPLEMENTATION AND AUDIT DETAILS

Readout and numerics. For the DomainNet experiment,

$$\begin{aligned} W_S &= (I + G_S)^{-1} \sum_{i \in S} H_i^\top Y_i, & p(x) &= \text{softmax}(h(x)^\top W_S), \\ b(S) &= \frac{1}{n_T} \sum_t \|p(x_t) - e_{y_t}\|_2^2. \end{aligned} \tag{12}$$

Y_i is the one-hot source-label matrix over the fixed 100-class vocabulary. There is no intercept or temperature tuning. Linear algebra uses float64 after loading float32 features. For effective rank obtained through a Gram eigendecomposition, negative roundoff eigenvalues are clipped to zero, square roots form singular values, and only $\sigma_j > \sigma_{\max} \sqrt{8\epsilon_{64} \max(n_S, d, 1)}$ are retained. The zero matrix returns rank zero in code. This tolerance concerns numerical rank; the entropy definition itself is unchanged. The shared projection uses seed 20260911 and $d = 128$; no outcome-dependent dimension choice is used within the frozen DomainNet protocol.

Diversity comparators. Bayesian D-opt maximizes $\log \det(I + G_S)$. Subspace DPP takes each block’s top 16 Gram eigenvectors U_i , defines $K_{ij} = \|U_i^\top U_j\|_F^2 / 16$ with $K_{ii} = 1$, numerically projects the matrix to a PSD correlation matrix, and maximizes $\log \det(K_{S,S} + 10^{-8}I)$. Its kernel describes block subspaces, not individual-image features. Domain balance maximizes the number of represented domains minus 10^{-3} times the sum of squared domain block counts. Deterministic scores break ties lexicographically by combination identifier. These choices are fixed comparators; poor performance of this DPP construction is not a result about all determinantal selection methods.

Construction and access checks. DomainNet records image content hashes, sample identifiers, official split membership, role assignments, and class vocabulary. Archive and split checksums are validated before feature extraction. Disjoint role membership and duplicate-content checks address overlap within the materialized benchmark; they do not rule out semantic near-duplicates or pre-training contamination. All screen methods use the same source blocks, projection, target-selection rows, and candidate family. Feature extraction ignores labels, while benchmark materialization uses taxonomy metadata and stores labels for later authorized stages.

Screen, validation, and audit manifests bind configuration and upstream artifact hashes. Stage-specific loaders enforce label and split permissions. Validation actually evaluates each method or random repeat separately, rather than substituting previously observed test outcomes. The test audit covers all 6×455 combinations per encoder only after choices are frozen. The exhaustive oracle is an audit reference, not a method eligible for deployment. Random repeats and shared combinations are not extra statistical units.

Study and artifact map. The archive retains historical experiment IDs to identify results without confusing them with the paper’s scientific questions. The following paths are relative to the anonymous code repository:

- Shared-model and shift studies (E1): `results/target_conditioned/e1_synthetic/expanded_v3_controlled_rng/`.
- PACS/Office-Home comparison (E2): `results/target_conditioned/e2_data_set_selection/method_v1/`.
- Post-hoc headroom (E2a): `results/target_conditioned/e2_dataset_selection/oracle_headroom_v1/`.
- Equal-cost shortlist (E2b): `results/target_conditioned/e2b_fixed_cost_shortlist/domainnet_v1/`.
- Risk sketch sweep (E5): `results/target_conditioned/e5_sketch_certificate/expanded_v3_controlled_rng/`.

The DomainNet data manifest begins 1ba25059df70be4d, candidate-set ID 545fe86c8c9e8304, and summary ID 2d473cb9eb977e85; full SHA-256 identifiers

are in the artifacts. Configuration is `code/configs/target_conditioned_e2b/domainnet_v1.json`; stage implementation is `scripts/run_target_conditioned_e2b_shortlist.py`. The manuscript’s `build_assets.py` regenerates the primary table, both figures, and bilingual abstracts from the committed sources and records full input hashes in `generated/asset_manifest.json`. It does not fit new models or alter experiment gates.

The three original feature matrices are not duplicated in Git. Metadata record their exact file and checkpoint hashes, shapes, preprocessing, extraction code revision, and runtime. Re-extraction requires access to the original public data and model weights; bitwise identity across different hardware is not assumed. The repository contains manifests and sufficient outcome tables to reproduce reported aggregate results without those matrices.

C ADDITIONAL FAILURE DIAGNOSTICS

Spectral intervals are not risk intervals. The four-encoder, ten-construction-seed spectral sweep uses six overlap settings and four budgets, with 960 configurations per sketch rank. At ranks 10, 20, 50, relative exact merged-effective-rank gains over rank-only are respectively 2.09, 3.81, 4.94% for the Gram sketch and 4.90, 5.48, 5.68% for DPP. These gains are in the geometry objective, not classification accuracy. Its entropy-continuity bounds follow the type of bound in Audenaert (2007); all 2,880 selected-prefix intervals cover, but none of 7,200 evaluated greedy steps is certified. The archived source is `results/two_stage_controlled_rank_sweep_corrected_v3/`. These observations do not imply that every low-rank sketch must fail.

Target-risk sketches. The separate E5 study compares sketch payloads to packed symmetric float64 Grams, not inflated dense storage. Among full-grid strategies whose mean byte ratio is at most 25%, the best selected-set match to Full-Gram greedy is 15.42%, achieved by adaptive rank cap 2 with mean byte ratio 14.91%. Its pairwise certificate rate is 3.91% and fully certified selection rate zero. Adaptive fallback reaches full selected-set agreement but costs approximately 160% of packed-Gram bytes on average when previous transmissions are counted. This fails the joint decision-and-communication requirement. The reference is Full-Gram greedy, not the exhaustive global minimizer.

A weak candidate family can make low regret misleading. An earlier, different experiment trains source-supervised bottleneck adapters on 21 sources and audits seven targets with four encoders and five adapter seeds. Target-train validation uses three stratified five-fold partitions, with held-out target test accessed separately after selection freezing. Frozen identity reaches mean accuracy 0.83654, an untrained random adapter 0.77144, and the validation-oracle source-supervised adapter 0.76951. The last trails identity by 0.06703, with target-cluster 95% interval $[-0.09771, -0.04400]$. Small shortlist regret relative to this adapter family is therefore not downstream benefit. This is not a DomainNet experiment or evidence against all adaptation methods. Results reside in `results/two_stage_repeated_cv_utility_corrected_v3/`.

Unfinished challenges. The current evidence does not include an independent replication of the fixed-cost shortlist protocol on another dataset, per-example uncertainty for the DomainNet top-ten set, a calibrated readout sensitivity study, or a matched end-to-end timing comparison against MMD. NTK, LoRA, and nonlinear training extensions are not results of this paper. Proposed experiments must be frozen before new outcomes are accessed; further analyses of already opened test results remain exploratory. No future result is used to support the present abstract or conclusion.